Report Date:January 29, 2021 File:120

Report Number:163340

Town of Ladysmith

PO Box 220, 410 Esplanade
Ladysmith, BC
V9G 1A2

Dear Town of Ladysmith

Re: Non-compliance Advisory Letter, Permit 120, 245 A Oyster Cove Road, Ladysmith BC, Effluent

On January 25, 2021, Ministry of Environment and Climate Change Strategy, Environmental Protection Division staff conducted an inspection of your facility, Town of Ladysmith located at 245 A Oyster Cove Road, Ladysmith BC with authorization number 120 under the *Environmental Management Act*.

This Advisory, the alleged violations and the circumstances to which it refers will form part of the compliance history of Town of Ladysmith, and will be taken into account in the event of future non-compliance.

Please note that this authorization is considered to be out of compliance until such a time as it can be confirmed to meet the authorization requirements.

Inspection Details:

On January 25, 2021, Ministry of Environment and Climate Change Strategy (Ministry) Environmental Protection Officer Darren Stewart (Ministry Staff) conducted a planned office-based inspection of the Town of Ladysmith (Permittee) Municipal Sewerage System (Facility) located in Ladysmith, British Columbia to verify compliance with Permit 120 (Permit). The Permit authorizes the discharge of effluent from a municipal sewage system to Ladysmith Harbour. The Permit was first issued on March 12, 1965, and most recently amended on January 10, 1994.
The Town of Ladysmith is currently undergoing the process of updating their discharge authorization to an Operational Certificate issued under their approved Stage 3, Liquid Waste Management Plan

The period assessed within this inspection report is from January 1, 2020, to December 31, 2020 (inspection period) and included a review of the following information:
• Quarterly (Q) water quality data reports (WQ DATA) submitted by the town of Ladsmith to the Ministry during the inspection period (2020 Q1 - Q4) which includes daily flow rate records and effluent water quality data.
• 2020 Yearly Bypass Report for the Ladysmith Wastewater Treatment Plant submitted January 09, 2021.
• any non-compliance reports (NCRs) and Dangerous Goods Incident Reports (DGIRs) reported to the Ministry in 2020, i.e. "trigger events".

The scope of this inspection was focused on monitoring data review, completed for reporting purposes related to the annual Conditional Management Plan (CMP) which is to provide enhanced management of the molluscan shellfish harvest in waters adjacent to the Town of Ladysmith wastewater treatment facility.
Below are the authorization clauses that were assessed for compliance during this inspection, as well as the associated details/findings and any actions required.

Ministry of Environment and Climate Change Strategy

Compliance
Environmental Protection Division

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Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.1: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.1 The maximum rate at which effluent may be discharged through the sewage treatment plant is 6 100 m3/d; the annual average rate of discharge is 3 050 m3/d. Flow in excess of treatment plant capacity is to be discharged to the bypass outfall. The maximum rate at which effluent may be bypassed is 8 500 m3/d; the total annual volume which may be discharged to the bypass outfall is 228 000 m3.
Details/Findings:	A review of the 2020 WQ DATA confirmed that the Town of Ladysmith exceeded the maximum allowable rate of discharge on 19 occasions during the inspection period as follows: January 07, 2020 = 6541 m3 January 19, 2020 = 7448 m3 January 20, 2020 = 7083 m3 January 21, 2020 = 7420 m3 January 22, 2020 = 7696 m3 January 23, 2020 = 10950 m3 January 24, 2020 = 7361 m3 January 28, 2020 = 6875 m3 January 31, 2020 = 8974 m3 February 01, 2020 = 12255 m3 February 02, 2020 = 6282 m3 October 13, 2020 = 6318 m3 November 16, 2020 = 6258 m3 November 17, 2020 = 7404 m3 November 19, 2020 = 6970 m3 November 20, 2020 = 6275 m3 November 21, 2020 = 9276 m3 November 30, 2020 = 7742 m3 November 31. 2020 = 6871 m3 The annual average rate of discharge was calculated by Officer Stewart from discharge rates included in the quarterly reports. The 2020 annual average rate of discharge was 3,190.25 m3/d, exceeding the allowable maximum annual average specified in this section.
Compliance:	Out
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.2: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.2 The characteristics of the discharge shall not exceed: Sewage Treatment Plant Discharge: 5-day Biochemical Oxygen Demand: 130 mg/L. Total Suspended Solids: 130 mg/L. Overflow Bypass Discharge: Screened combined stormwater and sewage effluent.

Details/Findings:	A review of the 2020 WQ Data confirmed that there were no exceedances for the characteristics of the discharge throughout the inspection period. All reported values during the inspection period for 5-day Biochemical Oxygen Demand (BOD5) and Total Suspended Solids (TSS) were well below permitted limits.
Compliance:	In
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.3: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.3 The works authorized are influent screening, a Spiragester primary type treatment plant, chlorination and dechlorination facilities, autothermophilic aerobic sludge processing facilities, foul air scrubber, sludge dewatering facilities, overflow bypass, bypass screening and solids removal facilities, 300 mm and 710 mm diameter parallel outfalls extending 850 m from mean low water to a depth of 17.9 m below mean low water, and related appurtenances approximately located as shown on the attached Site Plan A.
Details/Findings:	The authorized works were not evaluated as part of this office review inspection; therefore, compliance with this requirement was not determined.
Compliance:	Not Determined
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.1 Maintenance of Works and Emergency Procedures** 2.1: The Permittee shall inspect the pollution control works regularly and maintain them in good working order. In the event of an emergency or condition beyond the control of the Permittee which prevents continuing operation of the approved method of pollution control, the Permittee shall immediately notify the Regional Waste Manager and take appropriate remedial action.
Details/Findings:	On June 23, 2020 (DGIR 201029) the Town of Ladysmith reported a series of failures on their Salsnes primary filters, and were forced to run on only one out of the two filters after which the filter mesh on the only operable Salsnes filter ripped, causing a catastrophic failure. The report also specified that after this second filter failure, the plant flow could no longer reach secondary treatment and effluent was flowing into the spirogestor (after primary treatment). The spirogestor began flowing to the chlorine contact tank, where it received chlorination and dechlorination prior to discharge through the outfall pipe. The Town of Ladysmith immediately notified the Ministry via the Emergency Management BC (EMBC) hotline and initiated the appropriate remedial action.

Compliance:	In
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.2 Bypasses** 2.2: The discharge of effluent which has bypassed the designated treatment works is prohibited unless the consent of the Regional Waste Manager is obtained and confirmed in writing.
Details/Findings:	The 2020 Summary of Bypasses at the Wastewater Treatment Plant report submitted by the Town of Ladysmith to the Ministry on January 09, 2021, detailed two bypasses that occurred during the inspection period. On January 31, 2020 (DGIR 193885) a significant rain event led to 638 m3 of effluent bypassing the designated treatment works. On June 23, 2020 (DGIR 201029) a series of failures on the Salsnes primary filters led to 6899 m3 of effluent bypassing the designated treatment works. This discharge of effluent which bypassed the designated treatment works did not receive the consent of the Regional Waste Manager (now Director).
Compliance:	Out
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.5 Outfall Inspection** 2.5: The Permittee shall conduct a dye test on the outfall lines (or inspect by another method approved by the Regional Waste Manager) every five years or as may otherwise be required by the Regional Waste Manager.
Details/Findings:	The last dye test conducted on the Town of Ladysmith's wastewater treatment plant outfalls was completed on February 21, 2018; therefore, compliance with this requirement was not applicable throughout the inspection period.
Compliance:	Not Applicable
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.6 Chlorination** 2.6: Maintain a chlorine residual (at the point of discharge or prior to dechlorination) between 0.1 and 1.0 mg/L at all times.

Details/Findings:	Chlorine residuals (at the point of discharge or prior to dechlorination) were maintained between 0.1 and 1.0 mg/L at all times as confirmed by Officer Stewart during the data review of 2020 Q1, Q3 and Q4 WQ DATA submitted by the Town of Ladysmith. Please note, chlorine residual levels for Q2 2020 (April, May, June 2020) were not included in this compliance evaluation due to reported missing logsheets for that quarter regarding chlorine residuals and sodium hypochlorite usage.
Compliance:	In
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.11 Facility Classification** 2.11: The Permittee shall classify the wastewater treatment facility authorized in Section 1.1.3 (the facility) and the classification shall be maintained with the "British Columbia Water and Wastewater Operators Certification Program Society" (BCWWOCPS).
Details/Findings:	The Town of Ladysmith wastewater facility (facility #65) is classified as a WWT III with the Environmental Operators Certification Program (EOCP) with a classification expiration date of August 01, 2022.
Compliance:	In

Requirement Description:	**2. GENERAL REQUIREMENTS, 2.12 Operator Certification** 2.12: If the facility is classified by the BCWWOCPS (the Program) at Level II or higher, the Permittee shall ensure that all operators of the facility shall be certified by the Program to a Class I level, at a minimum, by December 1, 1994. Operators in Training: The Permittee shall ensure that all operators in training (OIT) working at the facility classified by the BCWWOCPS at Level II or higher shall be required to successfully pass an OIT examination within three (3) months of commencement of employment at the facility. The OIT certificate shall be valid for fifteen (15) months from the date of issue. Prior to the expiry date of the OIT certificate, but not sooner than twelve (12) months from the date when the OIT commenced facility operation, the OIT shall successfully complete a Class I certification examination in order to continue to operate at the facility. Chief Operator: Level II or Higher: If the facility is classified by the BCWWOCPS at level II or higher, the Permittee shall designate at least one operator to be the "Chief Operator" of the facility by December 1, 1996. The "Chief Operator" shall be certified at a Class II level, at a minimum. After December 1, 1996, no person shall have "Direct Responsible Charge", as defined by the BCWWOCPS, of a municipal wastewater treatment facility classified at Level II or higher unless they possess a valid operator's certificate not more than one level below the classification level of the facility. Chief Operator Level III and IV: If the facility is classified by the BCWWOCPS at level III, the Permittee shall designate a "Chief Operator", certified at a Class III level by December 1, 1998. If the facility is classified by the BCWWOCPS at Level IV, the Permittee shall designate a "Chief Operator" certified at a Class IV level by December 1, 1998.
Details/Findings:	Michael Brown is listed as the Chief Operator and certified to an appropriate level by the EOCP.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.1 Sampling and Analysis** 3.1.1: The Permittee shall install a suitable sampling facility and obtain a composite sample of the effluent from the treatment plant effluent once every month. The sample is to consist of four grab samples taken over a two-hour period at maximum flow and mixed to form a single sample for subsequent analysis. Proper care should be taken in sampling, storing and transporting the sample to adequately control temperature and avoid contamination, breakage, etc. Obtain analyses of the sample for the following: 5-day Biochemical Oxygen Demand; Total Suspended Solids.
Details/Findings:	Following a review of the 2020 WQ Data, it was confirmed that sampling and analyses were performed for Biochemical Oxygen Demand and Total Suspended Solids as required. Fecal coliforms samples were also collected and analyzed on a monthly basis.

Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.2 Flow Measurement** 3.1.2: Provide and maintain suitable flow measuring devices and record once per day the effluent volume discharged over a 24-hour period from the sewage treatment plant and the overflow bypass.
Details/Findings:	Daily effluent discharge flow rates were obtained and recorded as required by this section.
Compliance:	In
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.4 Reporting** 3.4: Maintain data of analyses and flow measurements for inspection and every three months submit the data, suitably tabulated, to the Regional Waste Manager for the previous quarter. The first report is to be submitted by January 31, 1994. Based on the results of the monitoring program, the Permittee monitoring requirements may be extended or altered by the Regional Waste Manager.
Details/Findings:	Data of analyses and flow measurements were submitted to the Ministry every three months as required by this section.
Compliance:	In

Compliance History:
2020-03-31 IR 147158 Advisory: AUTHORIZED DISCHARGES 1.1.1
2019-01-28 IR 115167 Advisory: AUTHORIZED DISCHARGES 1.1.1, Bypasses 2.2
2018-01-11 IR 76276 Advisory: AUTHORIZED DISCHARGES 1.1.1; 1.1.2, Chlorination 2.6, Outfall Inspection 2.5
2017-03-02 IR 49542 Advisory: Operator Certification 2.12

The Ministry of Environment Compliance and Enforcement Policy and Procedure (C&E Policy) prescribes common requirements and procedures for all Ministry staff to ensure consistent and risk-based assessment and response to non-compliance. Using the Non-Compliance Decision Matrix, the compliance determination for this inspection has been assessed as **Level 1, Category B, Advisory**.

More information about Environmental Compliance, the Non-Compliance Decision Matrix, and reporting and data submission requirements can be found at the links below:
General compliance information:
www.gov.bc.ca/environmentalcompliance
Non-Compliance Decision Matrix information:
www.gov.bc.ca/environment/how-compliance-is-assessed
Reporting and data submission requirements (to be sent to EnvAuthorizationsReporting@gov.bc.ca):
www.gov.bc.ca/submit-waste-authorization-reports

Please be advised that this inspection report may be published on the provincial government website within 7 days.

If you have any questions about this letter, please contact the undersigned.

Yours truly,

Darren Stewart
Environmental Protection Officer

cc:

Attachments:

Deliver via:

Email: [X] **Fax:** [ ] **Mail:** [ ]
Registered Mail: [ ] **Hand Delivery:** [ ]

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Please note that sections of the permit, regulation or code of practice referenced in this inspection record are for guidance and are not the official version. Please refer to the original permit, regulation or code of practice.

To see the most up to date version of the regulations and codes of practices please visit
http://www.bclaws.ca

If you require a copy of the original permit, please contact the inspector noted on this inspection record.

It is also important to note that this inspection record does not necessarily reflect each requirement or condition of the authorization therefore compliance is noted only for the requirements or conditions listed in the inspection record.